

Question number	Answer	Additional guidance	Mark
2(a)(i)	B is the correct answer • A is not the correct answer as actin does not join bone to bone • C is not the correct answer as muscle does not join bone to bone • D is not the correct answer as tendon does not join bone to bone		Computer (1)

Question number	Answer	Mark
2(a)(ii)	D is the correct answer • A is not the correct answer as actin does not join muscle to bone • B is not the correct answer as ligament does not join muscle to bone • C is not the correct answer as myosin does not join muscle to bone	Computer (1)

Question number	Answer	Mark
2(a)(iii)	D is the correct answer • A is not the correct answer as actin and myosin are not made of carbohydrates • B is not the correct answer as actin and myosin are not made of fatty acids • C is not the correct answer as actin and myosin are not made of nucleic acids	Computer (1)

Question number	Answer	Additional guidance	Mark
2(b)(i)	An explanation that includes two of the following points: • muscles are in antagonistic pairs / have extensor and flexor muscles (1) • the muscles contract (with equal strength) (1) • ligaments hold the joint in place (1)	muscles if named - rectus femoris and gastrocnemius / fibialis accept muscles work antagonistically ignore one contracts and the other relaxes accept ligaments hold the bones in place	Expert (2)

Question number	Answer	Additional guidance	Mark
2(b)(ii)	A description that includes three of the following points: • ATP binds to the myosin head (1) • {releasing the myosin head from the actin strand/breaking the cross-bridge} (1) • ATP is hydrolysed (1) • (providing energy) to change the myosin head {shape / position}(1) • allowing myosin to bind to actin / form actin-myosin cross bridge (1)	steps need to be in logical order to gain credit ACCEPT ATP broken down into ADP and Pi Do not accept the power stroke	Expert (3)

